

VIVEK KUMAR MISHRA

Fellow, British Computer Society (FBCS) | IEEE Senior Member | Sr. Software Engineer, The Vanguard Group
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Federal Recognition: Research permanently archived in the **U.S. National Transportation Library** (ROSA P dot/61210 & dot/59030). Methodologies adopted by the **Florida Department of Transportation**. Recognized petitioner for **EB-2 National Interest Waiver (NIW)** and **EB-1A Extraordinary Ability**—formal U.S. government acknowledgment of contributions to critical national infrastructure and the AI ecosystem. Aligns with **Executive Order 14110** on Safe, Secure, and Trustworthy AI.

PROFESSIONAL PROFILE

Vivek Kumar Mishra is a Senior Software Engineer at **The Vanguard Group** (Aug 2026–present), one of the world’s largest investment management firms with \$9+ trillion AUM. A distinguished scholar-practitioner with **12+ years** of contributions spanning enterprise technology, U.S. federally funded research, and international academic scholarship, he holds the title of **Fellow of the British Computer Society (FBCS)**—the most prestigious grade conferred by the UK’s Chartered Institute for IT—and is an **IEEE Senior Member**, held by fewer than 8% of IEEE’s 460,000+ global members. Prior to Vanguard, he served as Founding Engineer at **JPMorgan Chase & Co.** (Oct 2024–May 2026) on systems serving 80+ million Americans, delivering \$100K+ in savings and a 43% performance improvement. He is also the recipient of the **2024 Global Recognition Award** from an independent awarding body. His federally funded research at **California State University, Long Beach** has been permanently archived by the U.S. Department of Transportation and adopted by the Florida DOT. He is the author of 9 peer-reviewed publications (48+ citations, h-index 4) and the textbook *Design Patterns for AI Agents* (ISBN: 979-8268912371), adopted into AI curricula at three independent AICTE-regulated engineering institutions. He serves as peer reviewer for **NeurIPS**, **ICLR**, and **ICML** (all CORE A*) and has been featured in publications reaching tens of millions of readers globally.

DISTINCTIONS & PROFESSIONAL RECOGNITION

Fellow, BCS (FBCS)	Elected Fellow of the British Computer Society—The Chartered Institute for IT. Highest grade of membership; awarded for outstanding contribution to IT and exceptional professional leadership. (2025)
IEEE Senior Member	Top 8% of 460,000+ global IEEE members. Requires 10+ years practice and 5 years of significant performance, evaluated by independent Senior Members/Fellows. (2025)
Global Recognition Award	Recipient of the 2024 Global Recognition Award , presented by an independent global awarding body in recognition of outstanding professional achievements and contributions. (2024)
Innovation Awards	Winner, #FutureOfMeterReading Contest — Capgemini Super Techies (global competition). Global Project of the Year Award — Cognizant Technology Solutions (Fortune 200).

CAREER

The Vanguard Group Sr. Software Engineer, Aug 2026 – Present Malvern, PA	Architecting scalable, high-availability backend systems and AI-integrated data pipelines for one of the world’s largest investment managers (\$9+ trillion AUM), serving millions of retail and institutional investors globally.
JPMorgan Chase & Co. Sr. Software Engineer Oct 2024 – May 2026 Wilmington, DE	Founding Engineer, Chase Shopping Hub — platform serving 80+ million cardmembers , processing 5M+ monthly API requests at 99.9% uptime. Delivered \$100K+ in cost savings and 43% response-time improvement . Integrated LLM/ML pipelines into enterprise workflows.
CSULB — Research Asst. 2020–2021 Long Beach, CA	Hand-selected for U.S. DOT & Caltrans-funded Intelligent Transportation Systems research. Developed AI/Python frameworks adopted by Florida DOT across state lines. Authored 7 publications, permanently archived in the U.S. National Transportation Library (ROSA P). Contributes to federal Justice40 Initiative objectives.
Oracle Member of Technical Staff 2018–2020 India	Oracle Cloud Infrastructure (OCI) — storage, compute, and networking modules. Salary 83% above market average. Infrastructure automation via Ansible, Go, and Python.
Cognizant Tech. Solutions Programmer Analyst 2014–2018 India	Automated 500+ test cases; reduced CI runtime from 4.5 to 2.5 hours, saving 500+ engineering hours annually. Fortune 200 Global Project of the Year Award.

SCHOLARSHIP & ACADEMIC CONTRIBUTIONS

Publications	9 peer-reviewed articles 48+ citations h-index 4 . Cited by researchers in the U.S., Italy, China, and Iran. Selected works: “ <i>Framework for Rail Transport Inequality Assessment...</i> ” (15 citations); “ <i>Evaluating Financing Mechanisms ... Grade Separation Projects</i> ” (12 citations); “ <i>AI-Driven Optimization of Backend & Cloud Infrastructure...</i> ” (IJARSCT, IF 8.2); “ <i>Integrating Solar PV into the Grid: AI Application</i> ” (MDPI Sustainability, IF 3.9). Two works permanently archived in the U.S. National Transportation Library (ROSA P).
Authored Book	<i>Design Patterns for AI Agents: A Structured Approach to Building Robust and Intelligent Systems</i> (ISBN: 979-8268912371). Adopted into the B.Tech CSE AI curriculum at Anand International College of Engineering and Supaul College of Engineering; library collection at LNCTS Bhopal (all AICTE-regulated institutions).
Peer Review	Reviewer: NeurIPS 2025 (CORE A*, Position Paper Track, 3 papers); ICLR 2025 (CORE A*, FM-Wild Workshop); ICML 2026 (CORE A*, AIWILD Workshop); GSP 2025, ACDSA 2026, ICECET 2026. Academic journals: <i>Virtus InterPress</i> (Risk Governance; Business Strategy Review); <i>Informatica</i> .
Expert Judging	IEEE Senior Member Application Review Panel (June 2025)—evaluated engineering qualifications of professionals globally. Certified hackathon judge: PennApps (Univ. of Pennsylvania), DragonHacks 2024, Claro 2024.

MEDIA COVERAGE & PUBLIC RECOGNITION

Featured in **TechBullion** (230K/mo, exclusive interview); profiled in **VoyageLA** (“*Rising Stars: Meet Vivek Mishra*”); quoted as IEEE expert in **TechTarget** (1.4M+ daily), **Dice.com** (2.4M/mo), and **VKTR.com** (7M+ readers). Recognized by **IEEE** as an “*IEEE Impact Creator*” on IEEE Transmitter and IEEE.tv; featured via IEEE’s official X/Twitter. Profiled in **Dainik Bhaskar** (India’s #1 Hindi newspaper, 63M readers) and **Rajasthan Patrika** (25M readers).

EDUCATION & CERTIFICATIONS

M.S., Computer Science CSULB, 2021	California State University, Long Beach (R2 Doctoral University). Focus: Applied AI/ML & Intelligent Transportation Systems.
B.Tech, Info. Technology WBUT, 2014	West Bengal University of Technology, Kolkata, India.
Certifications	AWS Certified Cloud Practitioner IEEE Senior Member Fellow of BCS (FBCS)

Full publication list, media documentation, IEEE elevation letter, federal archival records, and letters of endorsement from 7 distinguished professionals available on <https://vivekmishra02.github.io/>.